

EXPERIMENT-8

I-V CHARACTERISTICS OF DOUBLE GATE N-MOSFET

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Course :M.Tech VLSI &ES

AIM:

Design a double-gate n-MOSFET (nMOSFET) and simulate its transfer (I_d - V_{gs}) and output (I_d - V_{ds}) characteristics using Synopsys TCAD Sentaurus.

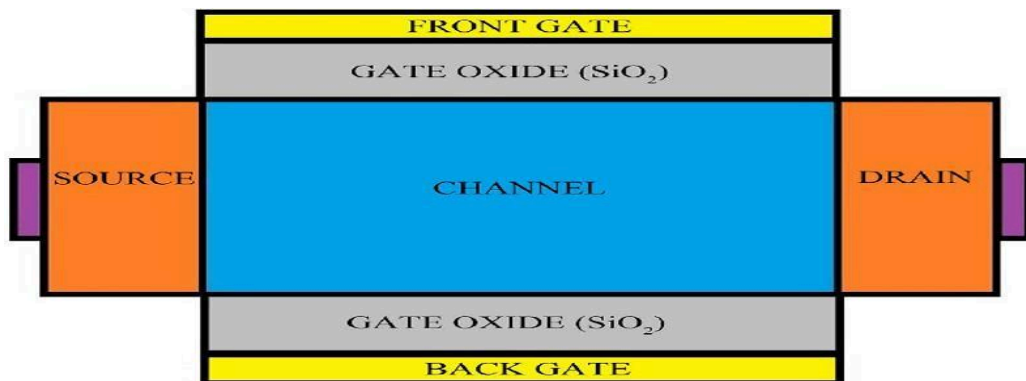
OBJECTIVES:

1. Study the I_d - V_{gs} characteristics of double gate n-MOSFET versus single gate n-MOSFET.
2. Study the I_d - V_{ds} characteristics of double gate n-MOSFET versus single gate n-MOSFET.

SOFTWARE TOOL:

Synopsys TCAD Sentaurus.

THEORY:



A MOSFET with two gates as opposed to one is called a double-gate N-MOSFET. A thin insulating layer divides the two gates, improving control over the channel region. Compared to the traditional MOSFET, the double-gate MOSFET offers several benefits, such as enhanced sub-threshold slope, decreased short-channel effects, and superior short-channel control. Additionally, the double-gate MOSFET may be readily scaled down to a channel length of 20 nm. The Silicon-on-Insulator (SOI) technique, which entails developing a silicon channel layer atop an oxide layer, can be used to construct the double-gate MOSFET. One of the most promising approaches to nanoscale integration is the double-gate MOSFET.

PROCEDURE:

- 1) Make a new folder. Launch a new folder in the Terminal, then type the command "sde" there.
- 2) Select 2D mode in GUI from view.
- 3) Choose "Exact Coordinates" from the Draw menu after deselecting "Auto Region Naming."
- 4) Click on a grid to inspect it, then adjust the grid height and width to 0.005.
- 5) Next, pick "Si" from the materials menu. Then, on the toolbar, click "create rectangular sheet region," which should be 180 nm long and 45 nm thick.
- 6) Position the rectangle onto the canvas and click "OK."
- 7) Type "silicon" as the region name now.
- 8) Next, create a 30 nm by 45 nm rectangle for the source region. Give it the source region name.
- 9) Additionally, create the same shape and use the same coordinates to label it the drain region.
- 10) Choose the silicon area for doping and use $1e+18$ boron concentration for doping. Phosphorus at concentrations of $1e+20$ and $1e+19$ is also used for doping in the drain and source regions.
- 11) go to the device's permanent profile location. Adjust the settings after choosing the silicon area.
- 12) Now select "sio2" as the material and place it near the edge of the Si semiconductor with a length of 120nm and 5nm thickness.
- 13) Take gate polysilicon and draw above on the oxide with the same dimensions.
- 14) Now draw the oxide and metal layers on top of the MOSFET with identical dimensions, and include the metal layer on the bottom as well.
- 15) To set Contact first change the select body to select an edge on the toolbar.
- 16) Now click on the arrow () placed on the left panel and go to contacts then place the electrical contact on the source, gate, source and body edges of the material.
- 17) Go to a mesh defined rel/eval window and select a rectangle draw a rectangle on canvas covering the entire design.
- 18) Go Mesh refinement placement enters the minimum & maximum element size. Click on Create Refinement.
- 19) Go to mesh a click on build mesh, then the device structure will be open in the

svisual.

- 20) To study the device characteristics, edit the diode “(cmd file).(msh.tdr file name should be included in the cmd file grid section.
- 21) Enter the “sdevicecmdfilename”
- 22) To visualize the device characteristics, enter svisual command in the terminal.
- 23) Go to file in svisual and open plot file.

RESULTS

Fig:1 Mesh for double gate NMOSFET

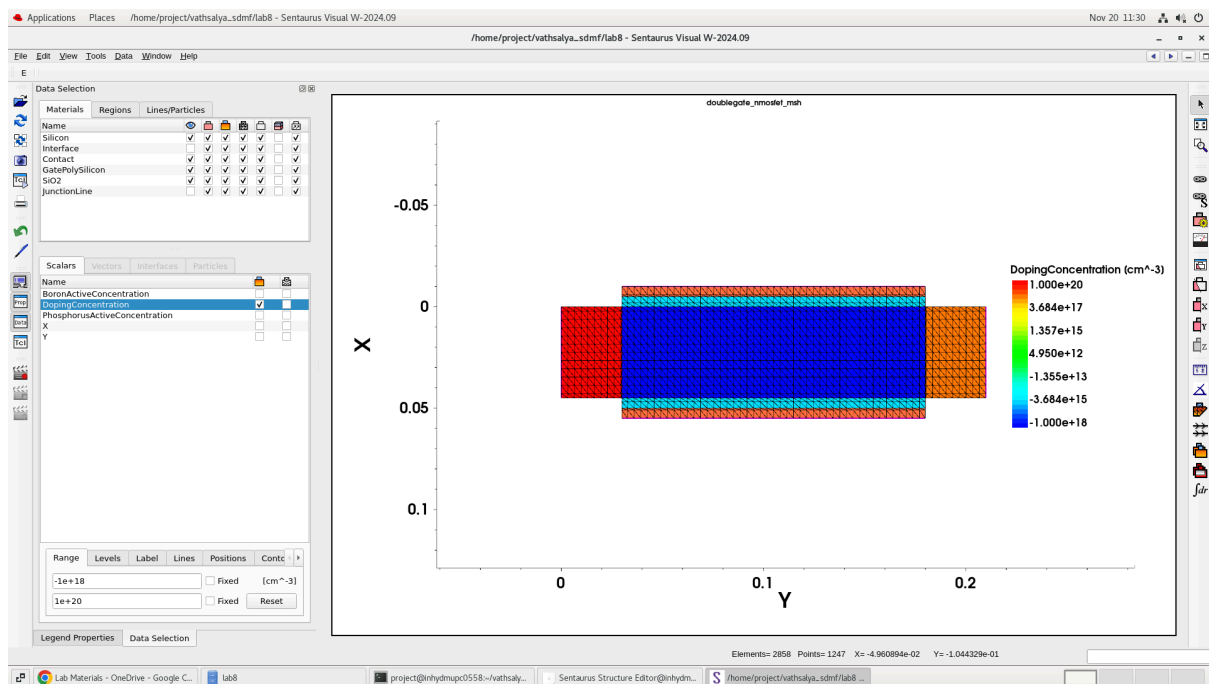


Fig.2a: Linear graph of I_d - V_{gs} Ion characteristics of double gate n-MOSFET vs single gate n-MOSFET

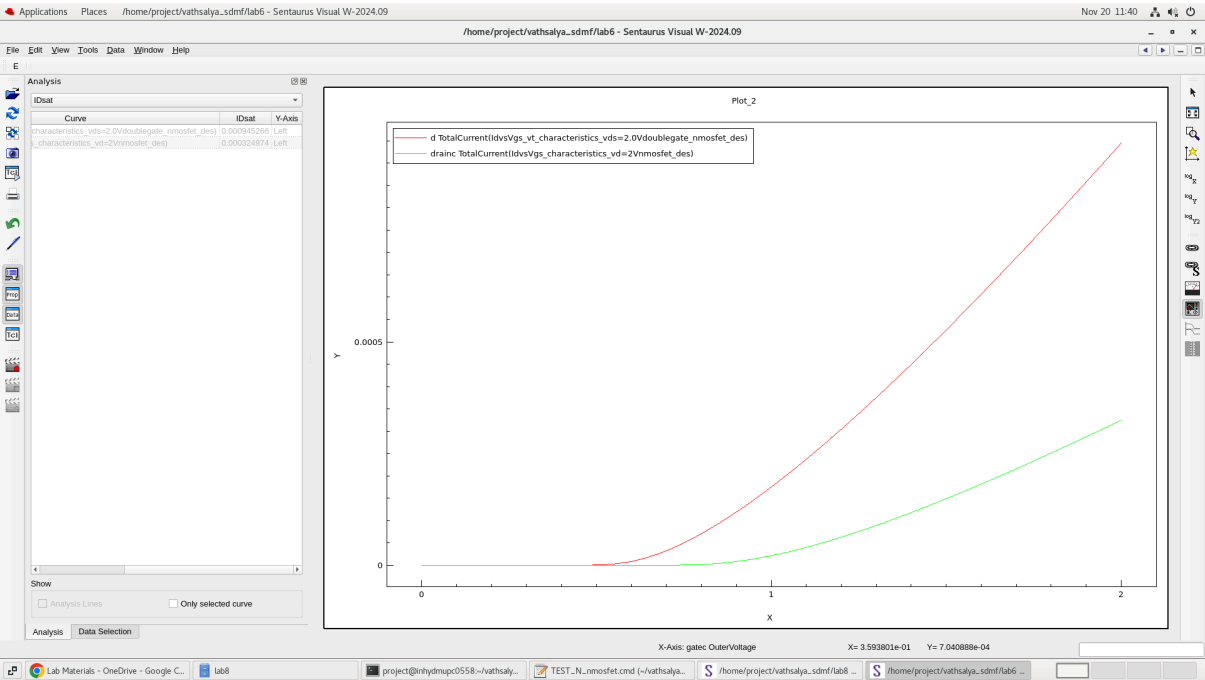


Fig.2b: Linear graph of I_d - V_{gs} Ioff characteristics of double gate n-MOSFET vs single gate n-MOSFET

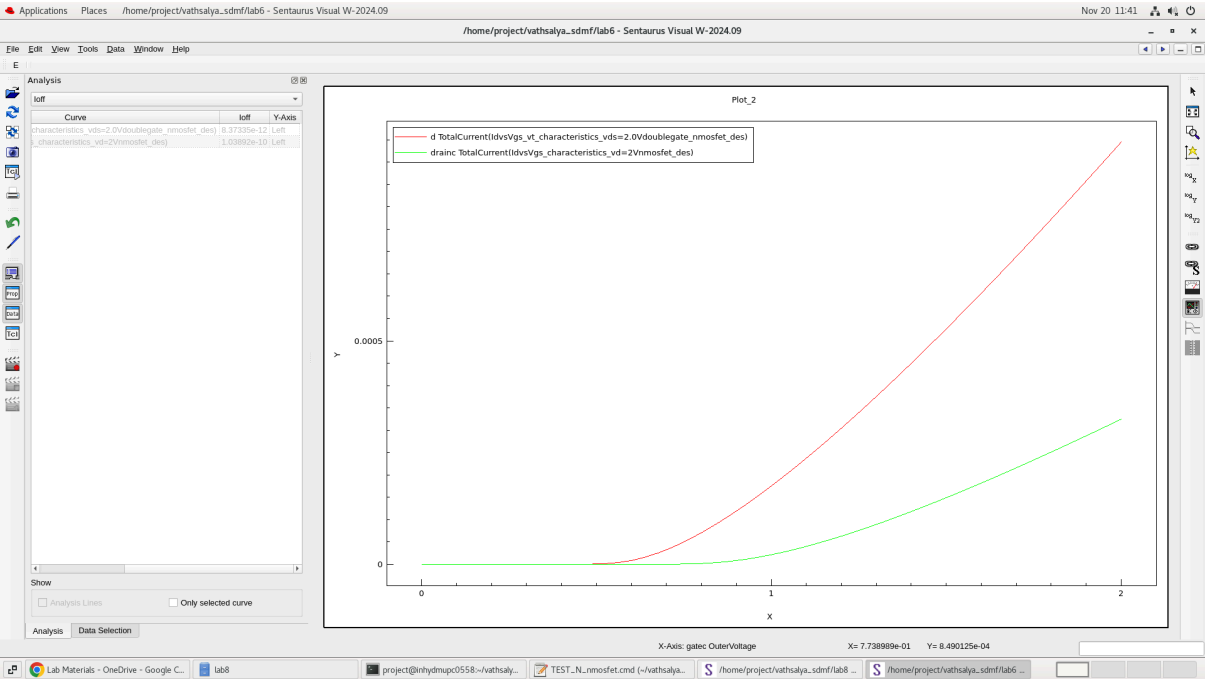


Fig.2c: Log graph of I_d – V_{gs} characteristics of double gate n-MOSFET vs single gate n-MOSFET.

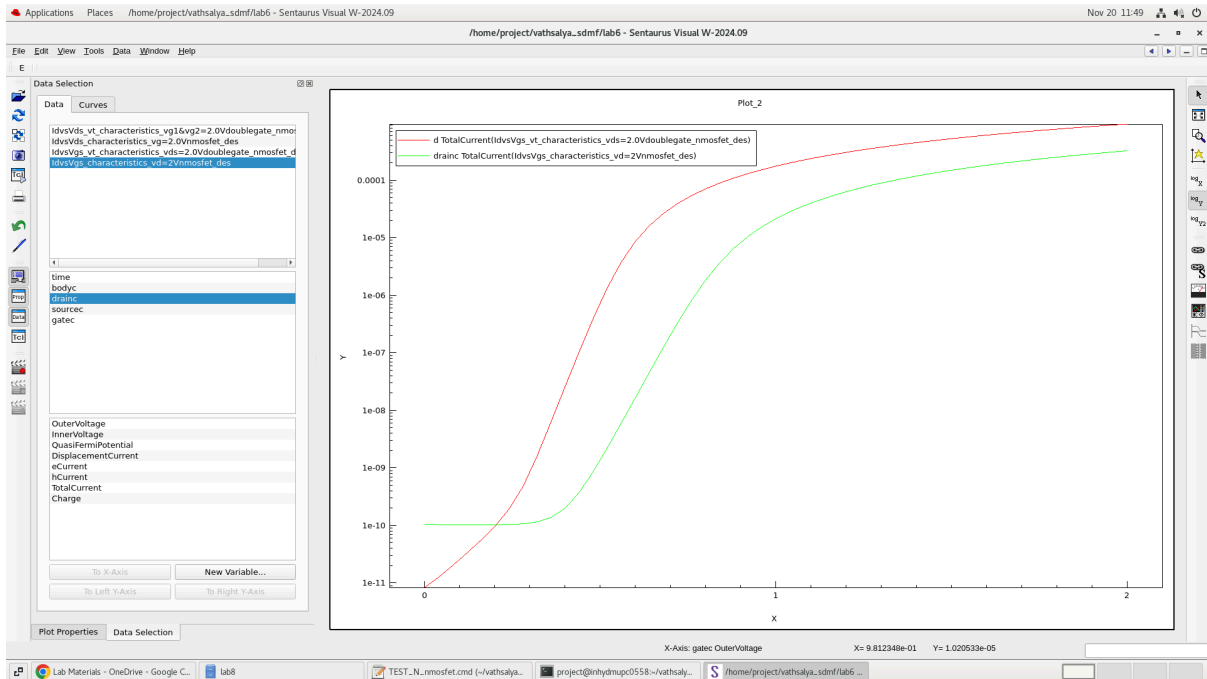
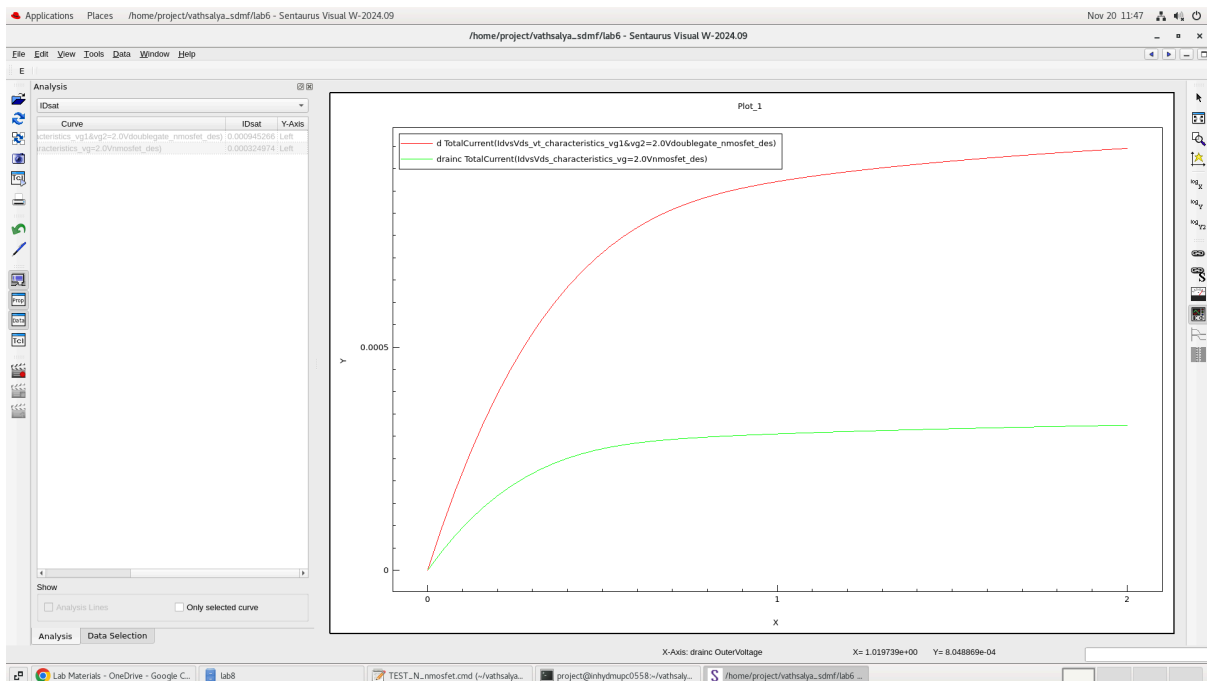


Fig.3: Linear graph of I_d – V_{ds} characteristics of double gate n-MOSFET vs single gate n-MOSFET.



OBSERVATIONS:

Double Gate vs Single Gate -> Transfer Characteristics

1. Double-gate MOSFET shows much higher I_d than the single-gate device for the same

V_{gs} .

2. Turn-on is earlier because DG gives better gate control → lower effective threshold voltage.
3. Subthreshold slope is steeper, meaning the DG device switches ON faster and with less leakage.

Double Gate vs Single Gate -> Output Characteristics

1. For the same V_{ds} , the double-gate MOSFET gives higher drain current (I_{dsat}) than the single-gate.
2. The DG device enters saturation smoothly with larger saturation current due to reduced channel resistance.

RESULTS:

The Transfer and Output Characteristics of the DG-MOSFET is studied in comparison with the single gate MOSFET.